

Normalised remainders for tight families of Taylor data

Claude, with Daniel Murfet

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Abstract

The localisation upgrade suggested by Astra #11. The bounded-family theorems of unit 150 required a deterministic bound $\|X_n\| \leq M$ on the random Taylor data, which excludes Gaussian-type limits. Here the hypothesis is replaced by tightness of the norms, $\forall \eta > 0 \exists M \forall^\infty n, \mu\{\|X_n\| > M\} \leq \eta$ (the content of the paper's appendix prop:convergence (i)). On the ball the error is deterministically small; off the ball the probability is at most η ; so the error tends to zero in probability (`tendstoInMeasure_zero_of_tight`), and Mathlib's `tendstoInDistribution_of_tendstoInMeasure_sub` yields $\text{normA} \Rightarrow A_\alpha(Z)$ and $\text{normB} \Rightarrow B_\alpha(Z)$ (`tendstoInDistribution_normA_tight`, `tendstoInDistribution_normB_tight`). Bounded families are tight (`tight_of_bounded`), so unit 150 is a special case. Zero sorry/axiom.

1 The things formalised

```
theorem tendstoInMeasure_zero_of_tight (f :  $\Omega \rightarrow \Omega \rightarrow$ ) (X :  $\Omega \rightarrow \Omega \rightarrow \text{CoeffPair}$ )
  (e :  $\Omega \rightarrow \Omega \rightarrow$ )
  (he :  $M, 0 \leq M \rightarrow \text{Tendsto} (e M) 1 (0))$ 
  (hf :  $M, 0 \leq M \rightarrow n \text{ in } 1, \quad , \|X n\| \leq M \rightarrow |f n| \leq e M n$ )
  (htight :  $\text{ENNReal}, 0 < \eta \rightarrow M, 0 \leq M \quad n \text{ in } 1, \quad \{ \|X n\| > M \} \leq \eta$ ) :
  TendstoInMeasure f 1 (fun _ => 0)
theorem tendstoInDistribution_normA_tight ... (X) (hXm :  $n, \text{Measurable} (X n)$ )
  (htight :  $\text{ENNReal}, 0 < \eta \rightarrow M, 0 \leq M \quad n \text{ in } 1, \quad \{ \|X n\| > M \} \leq \eta$ )
  (Z) (hX : TendstoInDistribution X 1 Z (fun _ => ) ' ) (Nseq)
  (hN : Tendsto Nseq 1 atTop)
  (h : polesBelowGen h h k k p p T) :
  TendstoInDistribution (fun n => normA ... (Nseq n) (X n)) 1
  (coeffA ... Z) (fun _ => ) '
theorem tendstoInDistribution_normB_tight ... : ... normB ... coeffB ... Z
theorem tight_of_bounded (X) (M) (hM :  $0 \leq M$ ) (hXb :  $n, \|X n\| \leq M$ ) : (tightness)
```

2 Mathematics

Fix $\varepsilon > 0$ and $\eta > 0$; choose M from tightness. For n large, $\text{errA}_M(N_n) < \varepsilon$, so $\{|err_n| \geq \varepsilon\} \subseteq \{\|X_n\| > M\}$ and $\mu\{|err_n| \geq \varepsilon\} \leq \eta$. Since η is arbitrary, $\text{err}_n \rightarrow 0$ in probability. The rest is the same as unit 150, with Mathlib's `tendstoInDistribution_of_tendstoInMeasure_sub` replacing the additive Slutsky lemma.

3 Paper-facing meaning

This is the form of the ordered normalised-remainder convergence in `thm:strataempiricalexpansion` that applies to the empirical process: the paper establishes tightness of $\tilde{\xi}_n$ from the moment bound $\mathbb{E} \sup |\tilde{\xi}_n|^s \leq C$ (appendix `lem:thm58new` and `prop:convergence (i)`); in our formulation the corresponding hypothesis is norm tightness of the Taylor data in `CoeffPair`, which follows from the paper’s bound by Cauchy estimates (not formalised) or can be taken as the standing hypothesis. Units 145–152 together give, for $d = 2$ and one chart: coefficient continuity, convergence in distribution of finitely many coefficients, convergence of the ordered normalised remainders for tight families, and exponential negligibility of the far region.

4 Lean notes

- `ENNReal.tendsto_nhds_zero`: $u \rightarrow 0$ iff $\forall \eta > 0, \forall^\infty n, u_n \leq \eta$, which is exactly the shape a tightness argument produces; `measure_mono` on the set inclusion finishes.
- `tendstoInDistribution_of_tendstoInMeasure_sub` takes the difference as the `Pi` subtraction $Y - X$; a term stated with an explicit `lambda` difference is accepted by `exact` (definitional unfolding of `Pi.sub_apply`).
- `zero_le` for `ENNReal` needs a type ascription `((zero_le : (0 : ENNReal) ))`; `zero_le _` fails since the argument is implicit.